Chapter 5

Applications of p-n junctions

5.1 Introduction

In the previous chapter, we have discussed about the basic properties of p-n junction diodes. From this discussion, it is evident that such a junction allows substantial current to flow through it only when it is biased in the forward direction. When reverse-biased, a very small current can pass through this diode. Thus, ideally, we can think of a p-n junction diode as *short-circuit* in forward bias and *open-circuit* in reverse bias. When an ac sinusoidal voltage is applied as input to a circuit consisting of a diode and a load resistance in series, a rectified signal is obtained at the output since the current can flow only in the forward direction. That is, the output signal across the load resistance contains only the positive half cycles of the original sine wave and hence has a positive average value. By suitably filtering this signal, ac to dc conversion can be achieved. This is perhaps the most common application of diodes.

Also, this property of a diode conducting only in the forward direction has been used for various switching applications. The time response of the diode may be a critical criterion for such applications. As already pointed out in Section 4.4.1, if a diode has to be switched on and off rapidly, the minority carrier lifetime must be small and therefore either deep levels have to be deliberately introduced during fabrication of the device or the device geometry has to be suitably adjusted so as to allow minimum charge storage.

Apart from their use as switching devices or rectifiers, specially designed p-n junction diodes are widely used in various other applications such as voltage regulator, variable capacitor, negative resistance devices and optoelectronic transducers. All these

applications are based on the fundamental junction properties. In this chapter, we shall discuss how the p-n junction can be suitably designed for some specific application.

5.2 Voltage Regulator

We have discussed the breakdown mechanisms of diodes in Section 4.5. It has been pointed out that the breakdown voltage of a diode can be varied by adjusting the doping concentration. When a diode is designed to have a specific breakdown voltage, it is called a **breakdown diode** or a **Zener diode**. The second term, however, is a misnomer in many cases, since the actual breakdown is usually due to avalanche multiplication of carriers. Zener effect dominates only when both sides of the junction are heavily doped.

Such a diode with a tailored breakdown voltage of V_{BR} can be used as a voltage regulator as shown in the circuit of Fig.5.1. Then, so long as the input voltage is greater than V_{BR}, the output voltage across the diode will remain fixed at V_{BR} even if there is a change in the value of the input voltage. In other words, a regulated voltage is obtained. This fixed breakdown voltage can also be used as a reference voltage in circuits that require a specific known value of voltage.

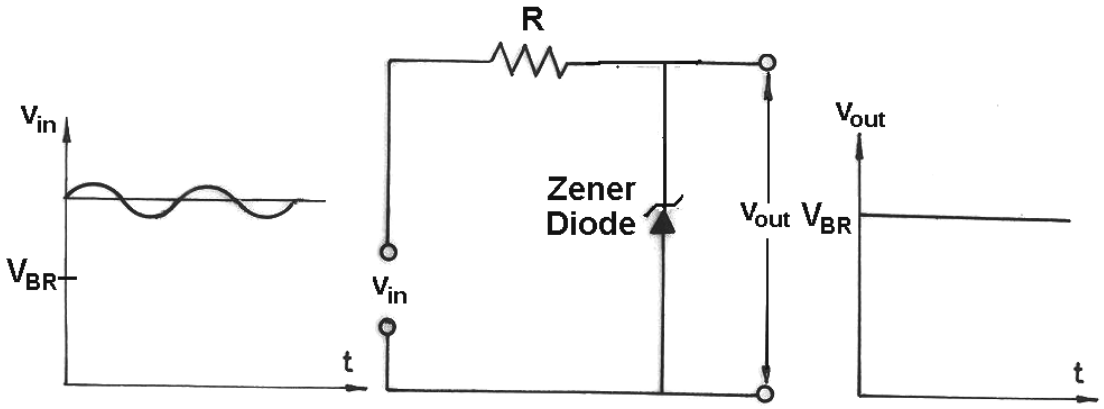


Fig. 5.1: A Voltage regulator circuit and the regulated output voltage signal

5.3 Variable Capacitor (Varactor)

Varactor is actually a short form of VARiable reACTOR. The property of a p-n junction that the width of the junction depletion region (and hence the depletion capacitance) is a function of the applied voltage is utilized in this application. As already pointed out in Section 4.2.2, the manner in which the depletion layer width changes with the applied reverse voltage will depend on the actual doping profile of the junction.

In general, let us consider a p^+n junction where the doping profile in the n-region is given by

$$N_D(x) = N_{DO}\left(\frac{x}{x_o}\right)^n \quad \text{.....(5.1)}$$

Then, following the usual procedure of integrating Poisson's equation twice using appropriate boundary conditions, the depletion width (W) as a function of the applied reverse bias (V_r) can be expressed as

$$W \propto (V_{bi} + V_r)^{1/(n+2)} \quad \text{.....(5.2)}$$

Consequently, using eqn. (4.33) and neglecting V_{bi} when a large reverse bias is applied, the depletion capacitance is

$$C_J = \frac{A\varepsilon_s}{W} \propto (V_r)^{-1/(n+2)} \quad \text{.....(5.3)}$$

For an abrupt junction, n = 0, and therefore the depletion capacitance C_J is proportional to $(V_r)^{-1/2}$ as seen from eqn. (4.31). Similarly, using eqn.(5.3), it can be shown that for a linearly graded junction (n = 1), C_J will be proportional to $(V_r)^{-1/3}$. So the voltage sensitivity $\left(\frac{dC_J}{dV}\right)$ is greater for an abrupt junction.

The sensitivity can be further enhanced by using a **hyperabrupt junction**, where the doping profile is tailored specifically to obtain a value of $n < 0$. A special case is seen for $n = -3/2$ is shown in Fig. 5.2, where, using eqn. (5.3), the depletion capacitance variation can be expressed as $C_J \propto V_r^{-2}$. If this varactor is used with an inductor (L) in a resonant circuit, the resonant frequency of the circuit will be given by

$$f_r = \frac{1}{2\pi\sqrt{LC}} \propto V_r \quad \ldots\ldots\ldots\ldots (5.4)$$

In other words, the resonant frequency of the circuit can be varied linearly by changing the applied reverse voltage. This property is widely used for tuning in radio/TV receivers.

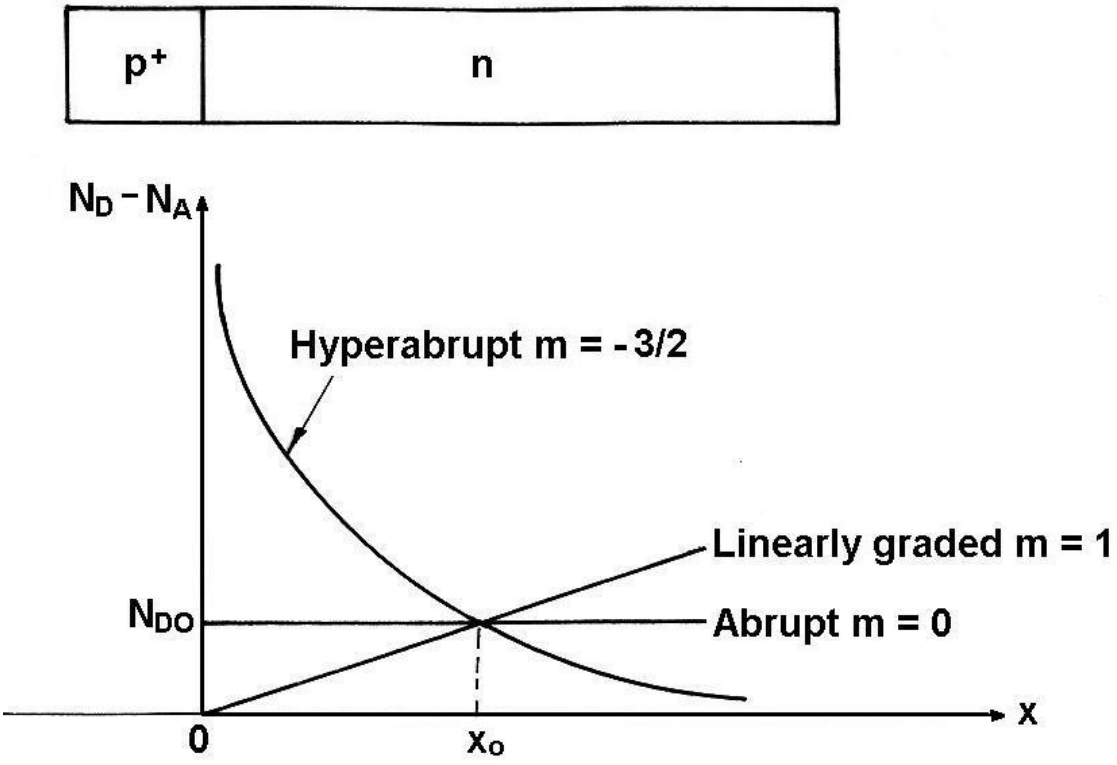


Fig.5.2 Hyperabrupt doping profile used in varactor diodes

5.4 Tunnel Diode

As the name itself suggests, this diode uses the principle of quantum mechanical tunneling of electrons through a potential barrier for its operation. This is essentially the **Zener effect** as discussed in Section 4.5.1, although very small reverse bias is needed in this case to initiate the Zener action. This device is also known an Esaki diode after L.Esaki, who received Nobel prize in 1973 for his work on this effect.

A significant difference between the tunnel diode and the other diodes discussed so far is that in the case of a tunnel diode a junction is formed between two **degenerate semiconductors**. By degenerate, we imply that the doping concentration in the semiconductor is so high that the interaction between the dopant atoms themselves can no longer be ignored. In such cases, the impurity levels cease to be discrete energy states and form a band instead. This band may overlap the bottom of the conduction band for donors and top of the valence band for acceptors. If the conduction band electron concentration is more than the effective density of states N_C, the Fermi level no longer lies within the energy gap but moves inside the conduction band. Similarly, if the hole concentration becomes greater than N_V, the Fermi level moves inside the valence band. The energy band diagram of a p-n junction at equilibrium where both p and n-regions are degenerately doped is shown in Fig. 5.3 (a). E_F is constant at equilibrium and lies below E_V on the p-side and above E_C on the n-side. When a small reverse voltage is applied, the quasi-Fermi level for holes (E_{Fp}) moves above that for electrons (E_{Fn}) as shown in Fig. 5.3(b). As there are filled states below E_{Fp} and empty states above E_{Fn} at the same energy and the barrier width is very small, electrons can tunnel from the p-side to the n-side. This is akin to Zener effect except that a very small reverse voltage is required to initiate

the tunneling process. As the reverse bias is increased, E_{Fp} moves further up with respect to E_{Fn}. There are now more filled states on the p-side at the same energy as empty states on the n-side. Thus the tunneling of electrons from p-side to n-side increases with increasing reverse bias.

On the other hand, when a small forward voltage is applied across this junction, the situation is depicted in Fig. 5.3(c). Now E_{Fn} moves up with respect to E_{Fp}, placing empty states on the p-side opposite to filled states on the n-side. Electron tunneling now occurs from n to p region and the tunneling increases with increase in applied voltage.

However, with increasing forward voltage, a point is reached where the conduction band edge on the n-side approaches the valence band edge on the p-side as shown in Fig. 5.3(d). The electrons on the n-side now see fewer accessible empty states on the p-side and hence tunneling reduces with further increase in the forward voltage. This is known as the negative resistance effect, since current decreases with increase in applied voltage. Once the conduction band edge on the n-side moves above the valence band edge on the p-side, tunneling stops and the device behaves like a conventional diode. The total current-voltage characteristic of a tunnel diode is shown in Fig. 5.4. It is observed that the current decreases rapidly beyond a voltage V_p, termed as the **peak voltage** of the tunnel diode. The ratio of the **peak current** (I_p) to the **valley current** (I_v) is often used as a figure of merit for this diode. The negative resistance region can be used for various applications such as oscillation, amplification and switching to name a few. Also, as tunneling does not have the time delays involved in drift and diffusion, tunnel diodes are very useful for high-speed circuits. However, the application of tunnel diode is limited by its comparatively low current drive.

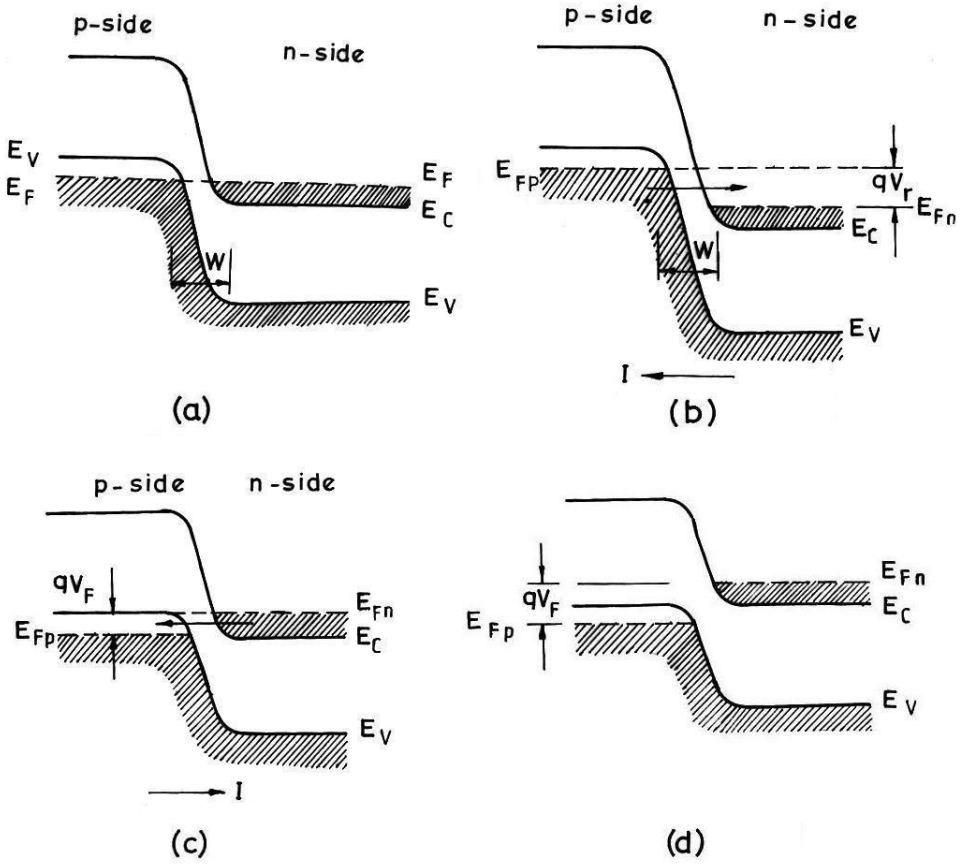


Fig.5.3: Energy band diagram of a Tunnel Diode under different bias conditions: (a) thermal equilibrium (b) reverse bias (c) small forward bias resulting in tunneling of carriers and (d) larger forward bias showing negative resistance effect

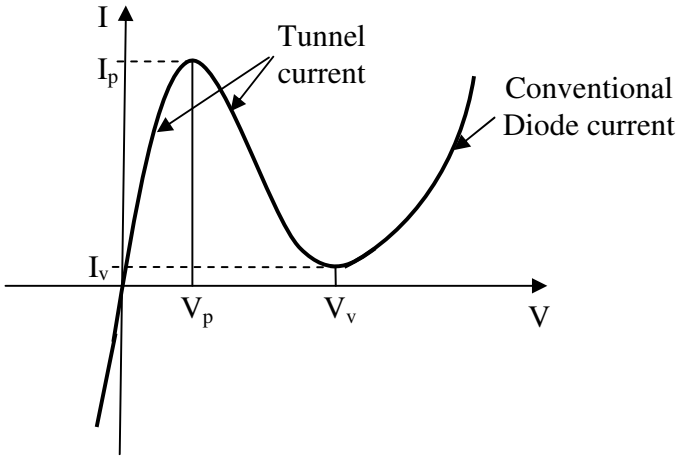


Fig.5.4: I-V characteristics of a Tunnel diode

5.5 Solar Cells and Photodiodes

Optoelectronic devices convert optical energy into electrical energy and vice versa. The mechanism of converting optical radiation into electrical energy is called the **photovoltaic effect**. Two important devices based on the photovoltaic effect are Solar Cell and Photodetector. The inverse of the photovoltaic effect is termed **electroluminescence**, i.e. the phenomenon of light emission when subjected to electrical signal. LEDs and Lasers are examples of this effect.

5.5.1 Photovoltaic effect

It has been discussed in Section 1.1.3 that if a photon with energy hν which is greater than or equal to the bandgap E_g is incident on a semiconductor, absorption of light can take place with consequent generation of an EHP. In other words, in order to generate EHPs in a particular semiconductor, the wavelength of the incident radiation must be lower than a characteristic value for the material, called the **cut-off wavelength (λ_c)**. The ability of the material to absorb light is also dependent on another parameter, called the **absorption co-efficient (α)**. α has the unit of 1/length and $(1/\alpha)$, called the **effective absorption length**, is essentially a measure of the material thickness within which the incident photon is absorbed by the semiconductor. In fact, the fraction of the incident radiation which is absorbed within a distance x from the surface is given by $\left[1-\exp(-\alpha x)\right]$. Therefore, if α is larger, more of the incident radiation would be absorbed within the distance x. Also, the fraction of the incident radiation absorbed within a distance $(1/\alpha)$ is $\left[1-\exp(-1)\right]$ = 0.632. An absorption coefficient of $10^4 cm^{-1}$ corresponds to an absorption length of 1μm, which means that 63.2% of the incident radiation is absorbed within 1μm from the surface. The absorption coefficients for different

semiconductors are plotted as a function of the photon energy in Fig.5.5. For indirect bandgap semiconductors, the value of α at a given photon energy is usually lower for materials with higher E_g, as can be seen from the plots for Ge and Si. However, direct bandgap materials (such as GaAs) have much larger values of α even though their E_g may be larger than indirect bandgap materials, as seen from the plots for GaAs and Si. This means that for a given photon energy, a much thinner layer of GaAs (than silicon) is needed to absorb the incident radiation.

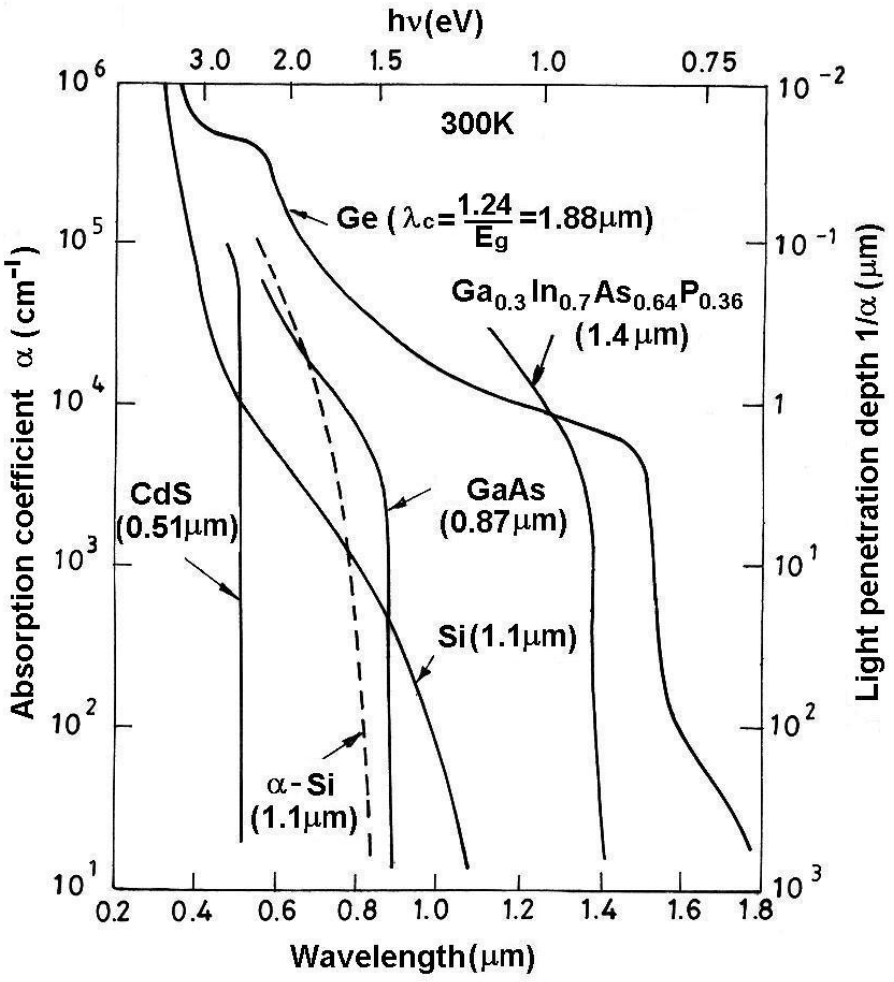


Fig.5.5: Absorption coefficients as functions of photon energy for various semiconductor materials[1]

When a p-n junction is illuminated with light of photon energy ≥ E_g, the photons are absorbed in the semiconductor and EHPs are generated both on the n-side and the p-side of the junction. For the EHPs to contribute towards current in the external circuit, the generated electrons and holes must be separated before they recombine. This is achieved if the EHPs are generated in the depletion layer, where the electric field sweeps away the electrons and holes in opposite directions. The light-generated minority carriers, which are generated within one diffusion length from the depletion layer edge, can also diffuse to the depletion region without recombining. They are then swept across the junction due to the electric field present in the depletion region as shown in Fig. 5.6. Due to the direction of the electric field being from the n-side to the p-side, the holes flow towards the p-side and electrons to the n-side. Since the direction of this light-generated current (I_L) is opposite to that in a forward-biased diode, we can write an expression for the total current in an illuminated diode as

$$I = I_0\left[\exp\left(\frac{V}{V_T}\right) - 1\right] - I_L \quad \ldots\ldots\ldots\ldots (5.5)$$

In other words, the total current of the diode is lowered by an amount I_L when illuminated.

The I-V characteristics of an illuminated p-n junction diode are shown in Fig. 5.7. An interesting point to be noted is that the characteristic encompasses first, third and fourth quadrant. Depending on the intended application, the diode can be made to operate so that either power is delivered to the device (operating in first or third quadrant) or supplied by the device to the external circuit (operating in fourth quadrant). When the diode is used as a solar cell, it is made to operate in the fourth quadrant and work as a

battery. On the other hand, as a photodetector, it is usually operated in the third quadrant. However, though the underlying principle is the same, the actual structures are very different for these two devices. We shall now discuss the operation of these two devices in the next two sections.

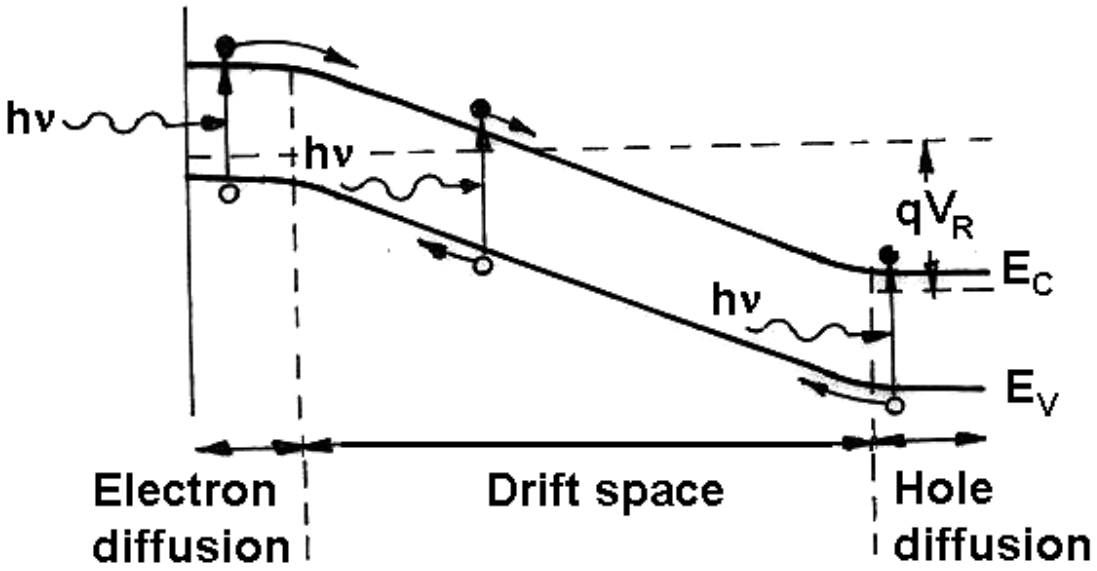


Fig.5.6: An illuminated p-n junction showing the generation and subsequent separation of EHPs

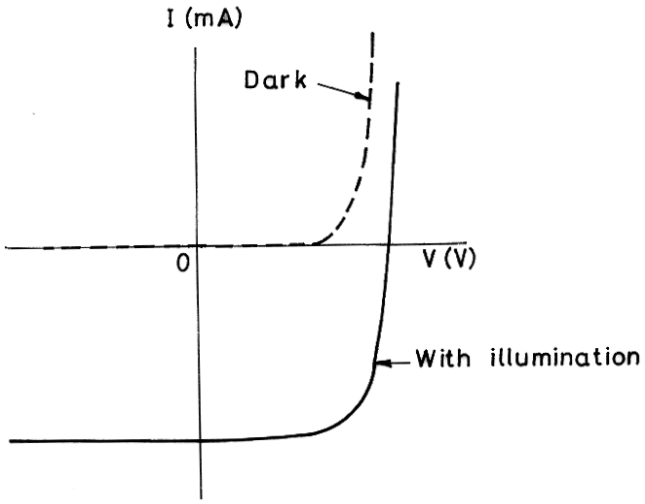


Fig.5.7 : I-V Characteristics of an illuminated p-n junction

Example 5.1 The bandgap of Silicon and GaAs are E_g = 1.12eV and E_g = 1.424eV respectively. What are the cut-off wavelengths (λ_c) in these materials? (Consider Planck constant (h) = $6.62\text{x}10^{-34}$ J-s and velocity of light (c) = $3\text{x}10^{10}$cm/s)

Solution : At the cut-off wavelength λ_c,

$h\nu = E_g$, or $(hc/\lambda_c) = E_g$, or $\lambda_c = (hc/E_g)$

Since 1eV = $1.6\text{x}10^{-19}$J, $hc = \dfrac{6.62\text{x}10^{-34}\,\text{x}3\text{x}10^{10}}{1.6\text{x}10^{-19}} = 1.24\text{x}10^{-4}\,\text{eV}-\text{cm} = 1.24\text{eV}-\mu\text{m}$

Therefore, when λ_c is expressed in μm and E_g in eV, we have the simple relation

$$\lambda_c[\mu\text{m}] = \frac{1.24}{E_g[\text{eV}]}$$

Therefore, for silicon, $\lambda_c = \dfrac{1.24}{1.12} = 1.11\,\mu\text{m}$

and for GaAs, $\lambda_c = \dfrac{1.24}{1.424} = 0.87\,\mu\text{m}$

5.5.2 Solar Cell

A solar cell is basically a p-n junction, which can deliver power to the external circuit when illuminated. Arrays of such junctions are used to supply electrical power to space satellites and also for remote and rural areas. Solar cells are preferred over conventional batteries for these applications as they are environment-friendly and provide maintenance-free service for long time. However, the high cost of solar cells has so far limited its widespread use for terrestrial applications.

Solar cells are usually large-area devices, so that a greater amount of optical energy is absorbed and the output power is high. In order to increase I_L, most of the

minority carriers generated outside the depletion region, must be able to diffuse to the edge of this region. This requires the diffusion lengths L_n and L_p to be large. As seen from the fabrication of diodes in Section 4.6, the top layer is produced by diffusion or ion-implantation with a high impurity concentration. High impurity concentrations create lattice damage which introduces recombination centres. This make the lifetime of carriers very short in the top layer and consequently the diffusion length is small. Hence, in a practical solar cell, the top layer is kept very thin so as to allow more light to reach the base. On the other hand, the diffusion length in the substrate (or Base) is large, and the carriers generated in this region mostly contribute to useful photocurrent. Also, n^+p structures, with the n^+ layer at the top, are preferred, as the diffusion constant of electrons in the p-base (D_n) is larger than that of holes in the n-base (D_p) of a p^+n junction. Larger diffusion constant means larger diffusion length and therefore an improvement in I_L, as more carriers generated outside the depletion region can diffuse to this high field region. Usually, in a practical solar cell the surface concentration of the n-region is kept below $10^{20}/cm^3$ with a junction depth of <0.2 μm. The surface of the solar cell is coated with **antireflection coating** in order to minimize reflection losses. Usual choices for antireflection material are SiO_2, TiO_2 and Ta_2O_5. The series resistance of the solar cell has to be made very small so that power is not lost due to losses in the cell itself. To achieve this, thin contact fingers are provided on the n-region. Fig. 5.8(a) shows the structure of a solar cell.

Solar cells are usually calibrated at AM1 or AM0 illumination. AM1 spectrum represents sunlight incident at the sea level on a clear day with the sun at the zenith and the incident optical power is about $92.5mW/cm^2$. AM0 is $135mW/cm^2$ and is considered

to be the intensity of the solar spectrum just outside the earth's atmosphere. While AM0 calibration is generally used for solar cells in space satellites, AM1 is usually meant for cells in terrestrial applications.

The fourth quadrant I-V characteristic of a solar cell is shown in Fig. 5.8(b). The three most important parameters of a solar cell are the **open circuit voltage (V_{oc})**, the **short circuit current (I_{sc})** and the **fill factor (FF)**. Since in a solar cell, current flows from the n-region to the p-region, i.e. in a direction opposite to that of a conventional diode, putting V = 0 in eqn.(5.5) we obtain the short circuit current (I_{sc}) as

$$I_{sc} = -I_L \ldots\ldots\ldots(5.6)$$

Again for I = 0, we have the open-circuit voltage (V_{oc}) from eqn.(5.5) as

$$V_{oc} = V_T \ln\left(1 + \frac{I_L}{I_o}\right) \approx V_T \ln\left(\frac{I_L}{I_o}\right) \ldots\ldots\ldots..(5.7)$$

However, no power is delivered when the solar cell is operating at either of these two points. So, the operating point of the cell should be selected such that the output power becomes maximum. The output power is given by

$$P = VI = V\left[I_L - I_o\left\{\exp\left(\frac{V}{V_T}\right) - 1\right\}\right] \ldots\ldots\ldots(5.8)$$

The condition for maximum power is obtained from dP/dV = 0. Hence differentiating eqn.(5.8) and equating it to zero, we get the expressions for the voltage and the current at maximum power point as

$$V_m = V_T \ln \frac{1 + \frac{I_L}{I_o}}{1 + \frac{V_m}{V_T}} = V_{oc} - V_T \ln\left(1 + \frac{V_m}{V_T}\right) \ldots\ldots\ldots(5.9a)$$

$$I_m = (I_o + I_L)\left[\frac{V_m}{V_m + V_T}\right] \approx I_L\left[1 - \frac{V_T}{V_m}\right] \ldots.(5.9b)$$

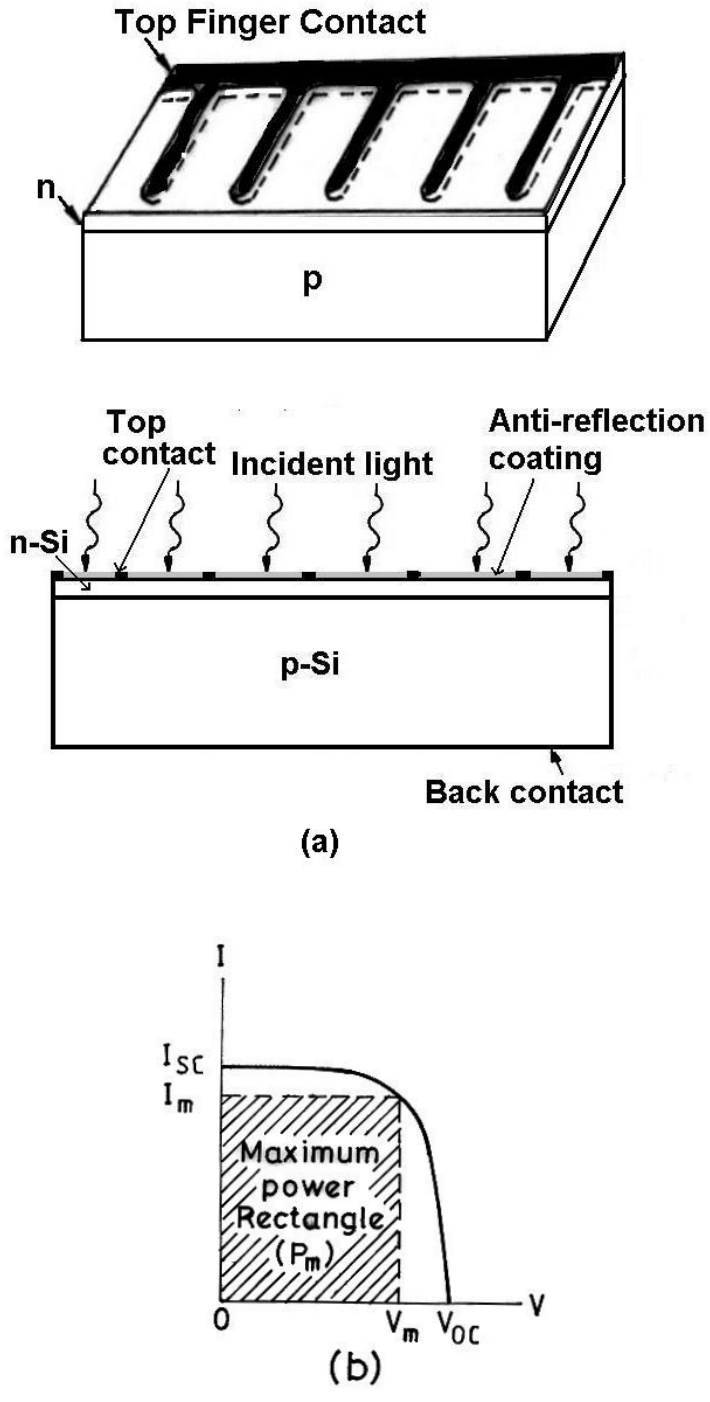


Fig.5.8 (a) Structure of a Solar Cell; (b): I-V characteristics of an illuminated Solar Cell showing the maximum power point

The optimum load resistance R_{op} for maximum power is thus given by rearranging eqn. 5.9(b) as

$$R_{op} = \frac{V_m}{I_m} = \frac{V_m + V_T}{I_o + I_L} \approx \frac{V_m}{I_L} \quad \text{.....(5.10)}$$

The fill factor (FF) of the solar cell is defined as

$$FF = \frac{V_m I_m}{V_{oc} I_L} = \frac{P_m}{V_{oc} I_L} \quad \text{.....(5.11)}$$

where $P_m = V_m I_m$ is the maximum power output of the solar cell. For a well–designed cell, FF is usually 0.7-0.8. The efficiency of the solar cell is given by

$$\eta = \frac{V_m I_m}{P_{in}} = FF \frac{V_{oc} I_L}{P_{in}} \quad \text{.....(5.12)}$$

where P_{in} is the incident optical power. Thus it is evident that to realize a solar cell with high efficiency, it is not only necessary to have high V_{oc} and I_{sc} but also a high FF. Solar cells with $\eta = 15\%$ under AM1 illumination are commercially available.

To improve the efficiency of solar cells, I_L has to be increased. This can be achieved by increasing the collection of photo-generated carriers. One way of improving the carrier collection is to dope the backside of the cell heavily so as to realize an n^+pp^+ structure. The potential barrier at the back pp^+ junction forces the minority carriers to be confined to the p-region. The decrease in the availability of minority carriers reduces recombination at the back contact and enhances the chance of collection of the generated carriers. Other techniques to increase I_L include surface texturing, use of solar concentrators, etc.

The efficiency of a solar cell depends, to some extent, on the material used. For a larger bandgap material, V_{oc} is usually higher as I_o reduces drastically with increase in E_g

(see eqn. (5.7)). On the other hand, with increase in E_g, some part of the incident light for which hν is now less than E_g, no longer contributes to I_L, thereby reducing I_{sc} (see eqn. (5.6)). There is therefore an optimum bandgap for the highest possible efficiency. GaAs solar cells have higher efficiency than their silicon counterparts, and are therefore preferred for space applications, where cost is not of much concern. On the other hand, silicon is still preferred for terrestrial applications, for their lower cost. Very high efficiency cascade solar cells, consisting of a series of junctions of materials of different bandgaps which efficiently absorb different portions of the solar spectrum, have also been demonstrated, primarily for solar applications.

Example 5.2 A solar cell has an area of 1cm x 1cm. The cell has an n^+p structure, where the doping concentration of the n-region is $10^{15}/cm^3$. Considering μ_n = 1350 cm^2/V-sec and τ_n = 1μs in the p-region, and I_L = 40mA, calculate I_{sc}, V_{oc} and maximum power output (P_m) if Fill Factor (FF) is 0.7. If the incident optical power is 90mW/cm^2, what is the efficiency (η) of the solar cell?

Solution : In the p-region,

$$L_n = \sqrt{D_n \tau_n} = \sqrt{\mu_n V_T \tau_n} = \sqrt{1350 \text{x} 0.026 \text{x} 10^{-6}} = 5.925 \text{x} 10^{-3} \text{ cm.}$$

Since a solar cell is usually a long base diode,

$$I_o = \frac{AqD_n n_i^2}{L_n N_A} = \frac{1.6 \text{x} 10^{-19} \text{ x} (1350 \text{x} 0.026) \text{x} (1.5 \text{x} 10^{10})^2}{5.925 \text{x} 10^{-3} \text{ x} 10^{15}} = 2.13 \text{x} 10^{-10} \text{ A}$$

Now, from eqn.(5.7), $V_{oc} \approx V_T \ln\left(\frac{I_L}{I_o}\right) = 0.026 \ln\left(\frac{40 \text{x} 10^{-3}}{2.13 \text{x} 10^{-10}}\right) = 0.495 \text{V}$

From eqn.(5.6), $I_{sc} = I_L = 40\text{mA}$

From eqn.(5.11), P_m = FF I_{sc} V_{oc} = 0.7 x ($40 \text{x} 10^{-3}$) x 0.495 = 13.86mW

From eqn.(5.12), $\eta = \frac{P_m}{P_{in}} = \frac{13.86}{90} = 0.154$

Therefore, the solar cell has an efficiency of 15.4%.

5.5.3 Photodiode

In a photodiode, the incident optical signal generates EHPs that gives rise to a photocurrent. The diode is reverse biased at less than its breakdown voltage ($V_r < V_{BR}$), so that under dark conditions, only a very small current flows, usually referred to as the **dark current** of the photodiode. Now, when light is incident on the diode, a large photocurrent flows. This change in current allows the photodiode to detect the presence of light and hence the photodiode is also termed a **photodetector**. The dark current must be made small and the photocurrent large as much as possible so that the photodiode can detect weak signals and the signal-to-noise ratio (SNR) is large.

Most photodiodes have a thin p^+ layer at the top, unlike solar cells where the n^+ layer is at the top. This is because a diode with the p^+ layer at the top has a faster time response compared to the diode with the n^+ layer at the top. (In a solar cell, response time is not important.) For a well designed photodiode, most EHPs are generated within the depletion region. Due to the electric field, the electrons move towards the n-region and the holes towards the p-region. Now the mobility of holes is much less compared to that of electrons. So with a p-layer at the top, the slower carriers (holes) have to move a much shorter distance to the p-contact.

An important parameter of a photodiode is the **quantum efficiency (η)**, which is defined as the number of useful EHPs generated per incident photon. By 'useful' EHPs, we refer to those which do not recombine and therefore contribute to current in the

external circuit. Now η may be less than 1 if all the light is not absorbed within the depletion layer or a diffusion length from it. Another related parameter of a photodiode is the **responsivity (R)**. It is the ratio of the photocurrent to the incident optical power and is expressed in units of Amperes/Watt. If P_{opt} is the incident optical power at a wavelength λ (corresponding to a photon energy $h\nu$), and I_L is the photo-generated current then η and R are expressed as

$$\eta = \frac{\text{rate of generation of useful EHPs}}{\text{no. of incident photons / unit time}} = \frac{\left(\frac{I_L}{q}\right)}{\left(\frac{P_{opt}}{h\nu}\right)} \approx \exp\left(-\alpha W_p\right) - \exp\left\{-\alpha\left(W + L_p\right)\right\}. \quad (5.13a)$$

$$R = \frac{I_L}{P_{opt}} = \frac{\eta q}{h\nu} \quad \ldots\ldots. (5.13b)$$

In eqn. (5.13a), α is the absorption coefficient, W_p is the thickness of the top p-layer, W is the depletion width and L_p is the minority carrier diffusion length in the underlying n-layer. It has been assumed that only those EHPs which are generated in the depletion layer and one diffusion length away from it on the n-side contribute to the photocurrent. From eqn.(5.13a), we see that for high efficiency, W_p must be small and the depletion layer width (W) must be large. Physically this means that most of the incident photons are absorbed within the depletion region rather than in the neutral p and n regions. As there is a strong electric field in the depletion region, as soon as an EHP is generated within this region, electrons are swept to the n-side and holes to the p-side and contribute to photocurrent. Also a large W results in a small capacitance and improves the speed of response of the diode. However, if W is too large, the transit time delay of the carriers will dominate and hence a compromise has to be made.

Q5.1 From eqn.(5.13b), we find that for a given quantum efficiency, the responsivity increases with wavelength. What is the physical reason for this?

Let us consider two ideal photodiodes, P_1 and P_2, each having quantum efficiency (η) equal to 1. Let us assume that the same number of photons is incident on P_1 and P_2 per unit time. Therefore, it follows that the light generated current is also the same in P_1 and P_2, i.e. $I_{L1} = I_{L2}$. Let λ_1 and λ_2 be the incident wavelengths on P_1 and P_2 respectively, and $\lambda_1 > \lambda_2$. Since each photon incident on P_2 has higher energy than those incident on P_1, the incident optical power on P_2 (P_{opt2}) is greater than the incident power on P_1(P_{opt1}).

Now, since $P_{opt2} > P_{opt1}$ and $I_{L2} = I_{L1}$, it follows that $R_1 > R_2$, where R_1 and R_2 are the responsivities of diodes P_1 and P_2 respectively. Therefore for two photodiodes having the same η, responsivity increases with increase in wavelength.

Also note that one photon can generate only one EHP. If the energy of the photon $h\nu > E_g$, the excess energy ($h\nu - E_g$) does not contribute towards I_L, but is lost as heat in the substrate. Therefore reduction in wavelength below the cut-off wavelength only increases the input power, without increasing output current, thereby reducing R.

A convenient way of tailoring the depletion layer width is to use a **p-i-n photodiode** rather than a simple p-n junction. Most of the reverse bias appears across the "intrinsic" layer, which need not be truly intrinsic, as long as the doping is low. As the doping is low, the carrier lifetime is high in this region and most of the photo-generated carriers are collected by the n and p-regions. Fig. 5.9(a) shows the cross-section of a p-i-n photodiode. The curvature at the junction edges leads to higher electric field intensity,

and a smaller radius of curvature results in a smaller breakdown voltage. A guard ring of larger junction depth is usually employed in p-i-n photodiodes as shown in Fig. 5.9(a) to increase the breakdown voltage and reduce the dark current.

Photodiodes, which are operated close to the avalanche breakdown voltage to enable avalanche multiplication of photo-generated carriers, are called **avalanche photodiodes**. The avalanche multiplication process increases the photocurrent and hence the responsivity of the detector. However, since the avalanche multiplication process is statistical in nature, there is a fluctuation in the responsivity due to the **avalanche multiplication noise**.

It is evident from our discussion in Section 5.5.1 that the photodiode is extremely sensitive to the wavelength of the incident light through the absorption co-efficient (α). If the incident photon energy $h\nu < E_g$, the light will not be absorbed. On the other hand, if $h\nu >> E_g$, α is large and the photons are absorbed very near the surface where the carrier recombination rate is very high. Therefore, the photocurrent will again be small. Hence, it is necessary to choose a proper photodiode material depending on the particular optical signal to be detected. **Heterojunction photodiodes**, with multilayers of different compound semiconductors are very useful in this regard. The bandgap of the absorbing layer can be tailored to match the wavelength of the incident light. Also, the quantum efficiency of a heterojunction photodiode is higher than a homojunction device. This is because, in a heterojunction diode, light can be made to reach the absorbing layer through a higher bandgap material (window), which does not absorb the incident radiation so that surface recombination is avoided. However, such a diode may have a higher dark current due to lattice mismatch at the heterointerface.

The $In_{0.53}Ga_{0.47}As/InP$ heterostructure is widely used for photodetector applications, since $In_{0.53}Ga_{0.47}As$ has a nearly perfect lattice match with InP. InP is used as the optical window and $In_{0.53}Ga_{0.47}As$ as the absorbing layer as shown in Fig.5.9(b). Since E_g of $In_{0.53}Ga_{0.47}As$ is 0.73eV, this photodiode can be used up to an optical wavelength of 1.7μm and is particularly suitable for optical communication in the range of 1.3 to 1.55 μm. At these two wavelengths, the optical fibre exhibits lowest attenuation and hence these wavelengths are deemed to be particularly suitable for optical communication.

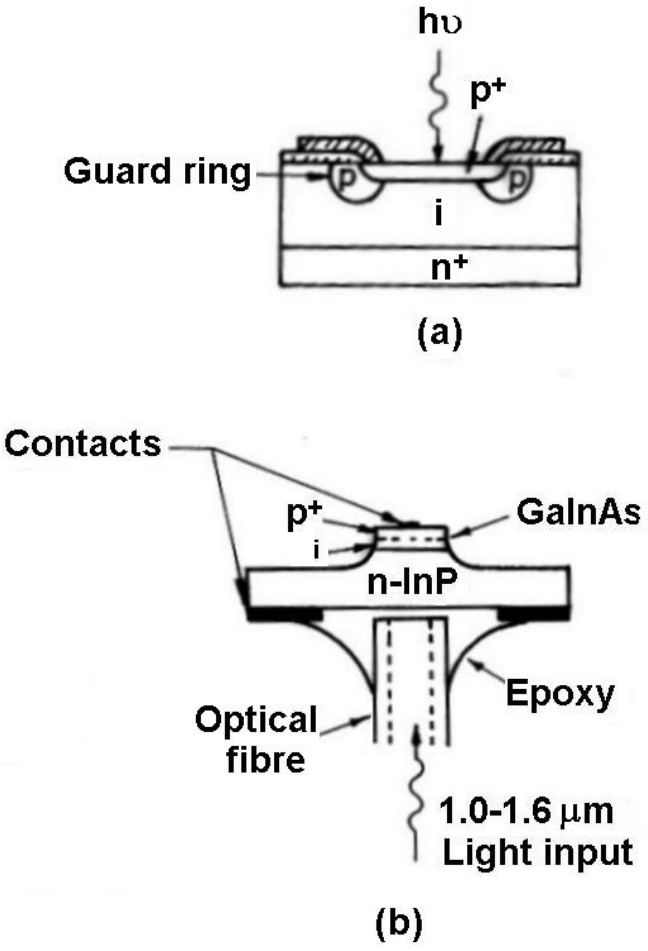


Fig.5.9 (a) Cross-section of a silicon p-i-n photodiode; (b) cross-section of a heterojunction InP/InGaAs p-i-n photodiode[2]

5.6 Light emitting diodes (LEDs) and Lasers

In this section we shall discuss those p-n junction devices, which convert electrical energy to optical energy. This effect is called injection **electroluminescence**. Though both LEDs and Lasers emit light when forward biased, Lasers emit *coherent* light in much narrower wavelength bands than LEDs and are more directional. This is because the dominant operating process for LEDs is **spontaneous emission** and for Lasers it is **stimulated emission**. We shall discuss these two processes in the following section.

5.6.1 Spontaneous and Stimulated Emission

We have already seen that a photon of appropriate energy can be absorbed by a semiconductor, creating an EHP in the process. This is called optical absorption. Let us consider Fig. 5.10(a) showing two energy levels in a semiconductor E_1 and E_2, where E_1 corresponds to the ground state and E_2 to the excited state. At room temperature, most of the electrons are in ground state. When a photon of energy greater than or equal to $h\nu_{12} = E_2-E_1$ is incident on the system, an electron in the ground state absorbs it and goes to the excited state. However, the excited state is unstable. So, after a short time, without any external stimulus, the electron comes back to the ground state giving off a photon of energy $h\nu_{12}$. This is called spontaneous emission and is schematically represented in Fig. 5.10(b). On the other hand, if a photon of energy $h\nu_{12}$ impinges on the electron while it is in the excited state, as shown in Fig.5.10(c), the electron can be stimulated back to the ground state with emission of a photon having an energy $h\nu_{12}$, in phase with the incident radiation. This is called stimulated emission. The radiation from stimulated radiation is monochromatic because each photon has energy precisely equal to $h\nu_{12}$ and coherent because they are all in phase.

5.6.2 Light Emitting diodes

LEDs are p-n junctions that can emit spontaneous radiation in ultraviolet, visible or infrared regions. Visible LEDs are widely used as visual information links between electronic instruments and their users. Thus it is a common sight to see them blinking on the control panel of almost all electronic gadgets. On the other hand, infrared LEDs are used in fibre-optic communication systems where silica fibres are used to guide the optical signal over a long distance. An important application of infrared LEDs is in opto-isolators, where light is generated when input electrical signal is applied to the LED. Subsequently, the light is detected by a photodiode and converted back to an electrical signal as a current flowing through a load resistor. Opto-isolators allow signal transmission at the speed of light. At the same time, the input and output electrical signals are electrically isolated.

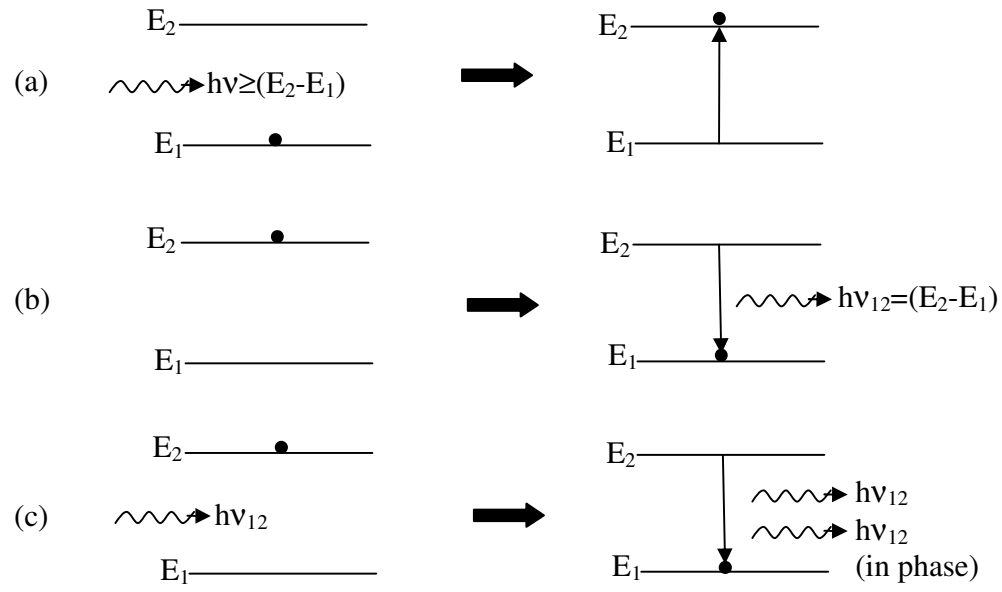


Fig.5.10 Schematic showing the three basic processes : (a) photon absorption; (b) spontaneous emission; (c) stimulated emission

$GaAs_{1-y}P_y$ is the most preferred material for visible LED. The phosphorus mole fraction in this ternary compound is denoted by y, i.e. as y is increased, more and more phosphorus atoms replace arsenic in the crystal lattice. For $0<y<0.45$, the bandgap of the material is direct and increases from 1.424eV (at y = 0) to 1.977eV (at y = 0.45). As already pointed out in Section 1.1.3, the probability of direct (radiative) transition is high in direct bandgap semiconductors. Hence $GaAs_{1-y}P_y$ ($y<0.45$) is used for light emission in the wavelength range of 627nm to 870nm. For $y>0.45$, the material has an indirect bandgap. So, special recombination centers have to be introduced to facilitate radiative recombination. Incorporation of nitrogen results in such a recombination center. It introduces an electron trap level very close to the bottom of the conduction band and greatly enhances the probability of radiative recombination.

In general, red LEDs are fabricated on GaAs substrates while orange, green and yellow LEDs are fabricated on GaP substrates on which a graded $GaAs_{1-y}P_y$ layer is grown by epitaxy. In optical communications, to take advantage of the 1.3 and 1.55 μm low-loss windows in optical fibres, AlGaAs or InGaAsP substrates are used.

At low forward voltage, the LED current is dominated by the nonradiative recombination current, due mostly to surface recombination. At higher forward voltages, the radiative diffusion current dominates and light is emitted as the injected minority carriers recombine with the majority carriers through a radiative recombination process. Finally, at very high forward voltages, the current is limited by the series resistance. Thus, incorporating the series resistance loss and considering our discussion in Section 4.3.4, we can write

$$I = I_d \exp\left(\frac{V - IR}{V_T}\right) + I_r \exp\left(\frac{V - IR}{2V_T}\right) \qquad \text{.....(5.14)}$$

where $I_d \exp\left(\frac{V - IR}{V_T}\right)$ is the radiative diffusion current, $I_r \exp\left(\frac{V - IR}{2V_T}\right)$ is the nonradiative recombination current and R is the series resistance of the diode. Evidently, to increase the power output of the LED, I_r and R must be reduced.

The electrical input signal to an LED (e.g. the applied voltage) is usually modulated at high frequencies. This in turn modulates the injected current in the LED. Parasitic elements (e.g. the depletion layer capacitance and series resistance) can cause a delay in the light output. The ultimate limit on the speed of response will depend on the carrier lifetime (τ). The cut-off frequency of an LED is given by[3]

$$f_c = \frac{1}{2\pi\tau} \quad \text{.....(5.15)}$$

5.6.3 Semiconductor Laser

Virtually all the semiconductor lasers use direct bandgap material. The dominant operating process for semiconductor laser is stimulated emission. If the stimulated emission is to dominate over the spontaneous emission, the photon field energy density must be very high and therefore an optical resonant cavity is to be used. In addition, if stimulated emission of photons has to dominate over the absorption of photons, we must have more electrons in the excited state than in the ground state. This is called **population inversion**, since under equilibrium the reverse is true. To achieve population inversion in a laser, let us consider a p-n junction formed between two degenerate semiconductors. The energy band diagram of such a junction under thermal equilibrium is shown in Fig. 5.11(a). When a forward voltage is applied, electrons are injected from the n-side and holes from the p-side into the space charge region. When a sufficiently large forward voltage is applied, high injection occurs, i.e. there is a large concentration

of electrons in the conduction band and holes in the valence band in the space charge region. This is the condition of population inversion as shown in Fig.5.11(b). For band-to-band transition, the minimum energy required is E_g. So, we can write the condition for population inversion as $qV_F = E_{Fn} - E_{Ep} > E_g$, where V_F is the applied forward bias.

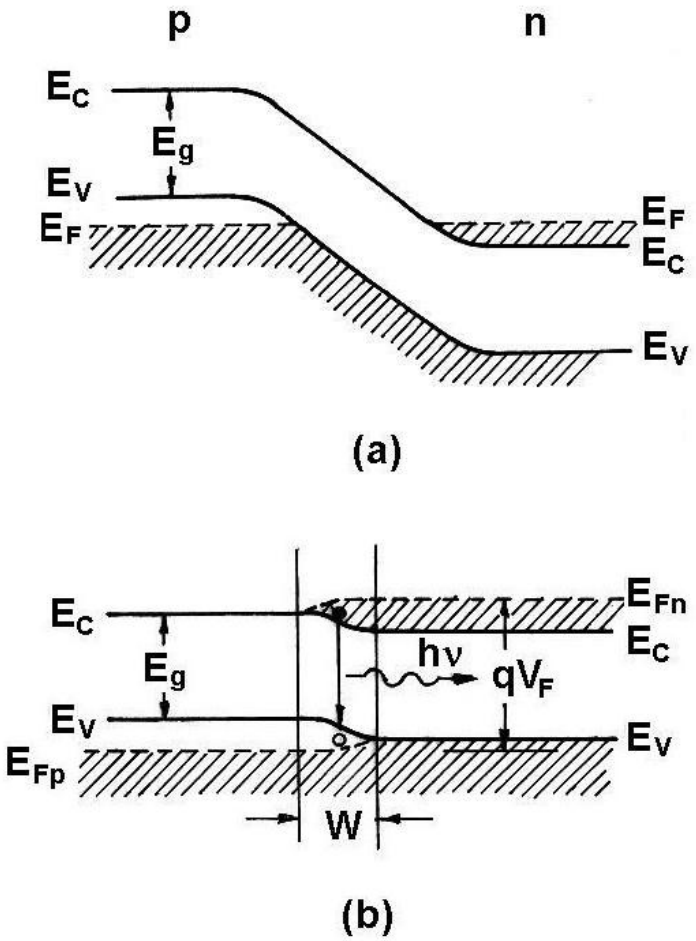


Fig. 5.11 (a) Energy band diagram of a laser diode junction under thermal equilibrium and (b) under large forward bias showing population inversion. The shaded regions show states which are occupied by electrons.

Figs. 5.12(a) and (b) show two commonly used laser structures. The first structure is a basic p-n homojunction laser that has the same material (e.g. GaAs) on both sides of the junction. A pair of parallel planes, which are perpendicular to the plane of the junction, are cleaved or polished. Under appropriate biasing condition, laser light is emitted from these planes. The other two sides are deliberately roughened to prevent lasing in those directions. This is called a Fabry-Perot resonant cavity with a typical cavity length of 300μm. The other structure, shown in Fig.5.12(b), is called a double-heterostructure (DH) laser, where a thin layer of narrow bandgap material (active region), e.g. GaAs is sandwiched between layers of a wider bandgap material, e.g. $Al_xGa_{1-x}As$. This is usually realized by epitaxy. In such a structure, the carriers are better confined in the active region due to the heterojunction barriers. Optical confinement is also better in DH lasers due to the abrupt reduction of refractive index outside the active layer. Since the refractive index in the active GaAs layer is larger than the refractive indices of the surrounding $Al_xGa_{1-x}As$ layers, the propagation of the electromagnetic radiation is confined in a direction parallel to the layer interfaces. The **threshold current density (J_{th})** defined as the minimum current density required for lasing, is lower for DH lasers compared to homojunction lasers. With increase in temperature, J_{th} for DH lasers increases at a slower rate than that for homojunction lasers and hence DH lasers are preferred for continuous operation at room temperature.

At low currents, where the spontaneous emission dominates, the laser spectrum has broad spectral distribution, and the full width at half maximum intensity may be about 10-50nm. As the current approaches threshold, the spectral distribution becomes narrower and above J_{th}, the spectral width is usually of the order of 0.1-1 nm.

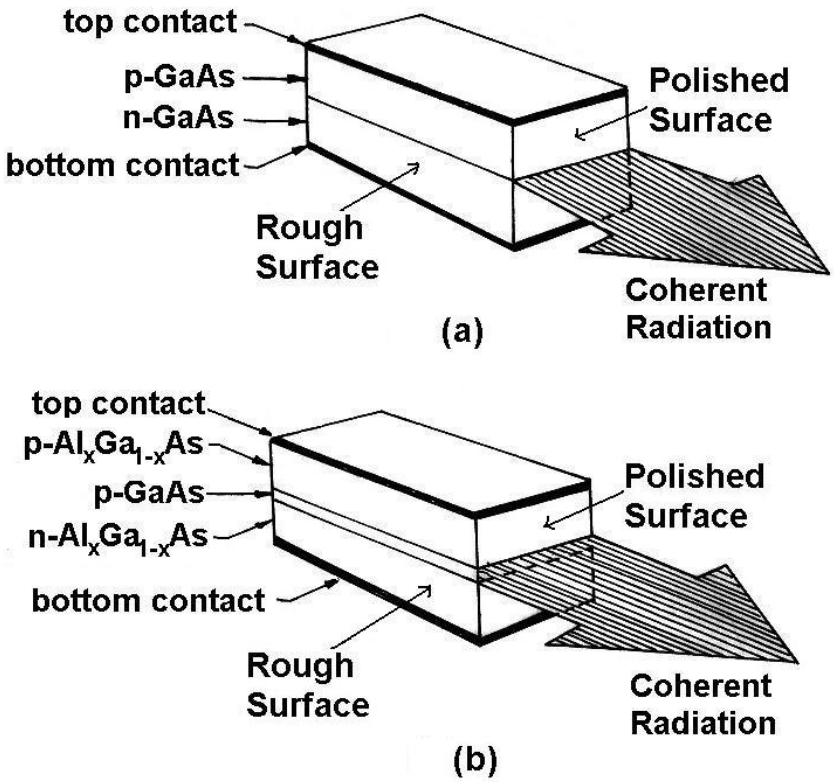


Fig.5.12 Structure of (a) p-n homojunction laser (b) Double Heterostructure (DH) laser

REFERENCES AND SUGGESTIONS FOR FURTHER READING

1. H. Melchior, “Demodulation and Photodetection Techniques”, in F.T. Arecchi and E.O. Schulz-Dubois, Eds., Laser Handbook, vol.1, North Holland, Amsterdam, pp.725-835, 1972.
2. T.P.Lee, C.A. Burrus and A.G.Dentai, “InGaAs/InP p-i-n photodiodes for lightwave communications at 0.95 to 1.65μm wavelengths”, IEEE J. Quantum Electron., QE-17, p.232, 1981.
3. S.M. Sze, Physics of Semiconductor Devices, 2nd ed., Wiley, N.Y., 1981.
4. J. Singh, Semiconductor Devices, New York: McGraw-Hill, 1994.
5. B.G. Streetman and S. Banerjee, Solid State Electronic Devices, 5th ed., Prentice Hall, New Jersey, 2000.
6. Bhattacharya P., Semiconductor Optoelectronic Devices, Englewood Cliffs, NJ: Prentice Hall, 1994.

PROBLEMS AND QUESTIONS

1. At a particular wavelength of incident radiation, the absorption coefficients of Si and GaAs are $4x10^3$/cm and $3x10^4$/cm respectively. What are the thicknesses of Si and GaAs necessary to absorb 80% of the incident radiation?

2. A 5μm thick silicon sample is illuminated with monochromatic light having $h\nu$ = 2eV. The incident power is 10mW. (a) What is the total energy absorbed in the sample per second? (b) How much energy per second is dissipated as heat? (c) How many EHPs are generated in the sample per second?